

# Reader chips

give me the datasheets and comparison for the following RFID reader chips. e710, e518, e310, r2000, r500

I’ll compare them as UHF RFID reader ICs/modules and I’ll verify the exact part names first, because “E518” may be a vendor shorthand or typo for Impinj E510/E710 family rather than a standard chip name.

Thought for 16s

Below is a practical datasheet-style comparison for the RFID reader chips you listed. One note first: I could not verify a standard “**E518**” Impinj reader chip. I think you may mean **E510**. Impinj’s current E-family reader chips are **E910, E710, E510, and E310**, and the public comparison tables list **E710 / E510 / E310 / R2000 / R500** together.

## Datasheet / product document links

| Chip                       | Datasheet / document                             |
|----------------------------|--------------------------------------------------|
| Impinj E710                | E910/E710/E510/E310 reader chip datasheet:       |
| Impinj E510                | E510 product brief: and full E-family datasheet: |
| Impinj E310                | Full E-family datasheet:                         |
| Impinj R2000               | R2000 product brief:                             |
| Impinj R500                | R500 product brief:                              |
| Portfolio comparison table | Impinj reader chip portfolio spec table:         |

## Main comparison table

| Item         | E710                     | E510 / likely “E518” | E310           | R2000             | R500              |
|--------------|--------------------------|----------------------|----------------|-------------------|-------------------|
| Generation   | Newer E-series           | Newer E-series       | Newer E-series | Older Indy series | Older Indy series |
| Air protocol | RAIN RFID / ISO 18000-63 | Same                 | Same           | Same              | Same              |

|                            |                                                |                        |                    |                                        |                                        |
|----------------------------|------------------------------------------------|------------------------|--------------------|----------------------------------------|----------------------------------------|
|                            | / EPCglobal<br>Gen2v2                          |                        |                    |                                        |                                        |
| Receive sensitivity        | <b>-88 dBm</b>                                 | <b>-82 dBm</b>         | <b>-75 dBm</b>     | <b>-84 dBm</b>                         | <b>-68 dBm</b>                         |
| Max read rate              | <b>~950–1000 tags/s</b>                        | <b>~550–700 tags/s</b> | <b>~250 tags/s</b> | <b>~900 tags/s</b>                     | <b>~190 tags/s</b>                     |
| Typical power consumption  | <b>~0.5 W</b>                                  | <b>~0.5 W</b>          | <b>~0.5 W</b>      | <b>~1.5 W</b>                          | <b>~1.1 W</b>                          |
| Package                    | 56-pin QFN                                     | 56-pin QFN             | 56-pin QFN         | 64-pin QFN                             | 64-pin QFN                             |
| Package size               | 6 × 6 mm                                       | 6 × 6 mm               | 6 × 6 mm           | 9 × 9 mm                               | 9 × 9 mm                               |
| Self-jammer cancellation   | Yes                                            | Yes                    | Yes                | Yes                                    | No / not listed in portfolio table     |
| Reader modes               | 11                                             | 9                      | 5                  | 4                                      | 4                                      |
| Integration level          | Radio + modem + MAC + baluns + power detectors | Same                   | Same               | Radio + modem                          | Radio + modem                          |
| Pin/software compatibility | Compatible with E710/E510/E310 family          | Same                   | Same               | Compatible with R2000/R500 family      | Same                                   |
| New design status          | Recommended                                    | Recommended            | Recommended        | <b>Not recommended for new designs</b> | <b>Not recommended for new designs</b> |

The strongest simple takeaway: **E710 is the best overall, E510 is the balanced middle choice, E310 is the low-cost/shorter-range choice**, and **R2000/R500 are older parts that Impinj marks as not recommended for new designs**.

## User-friendly explanation

### 1. Impinj E710

The **E710** is the best choice when you need long range, high sensitivity, and high tag population performance. It is suitable for fixed readers, handheld readers, shelves, cabinets, conveyors, gates, and industrial inventory systems.

Use **E710** when you want:

| Need                          | Good fit? |
|-------------------------------|-----------|
| Long read range               | Excellent |
| Many tags at once             | Excellent |
| High read speed               | Excellent |
| New product design            | Excellent |
| Low power compared with R2000 | Excellent |

**Best for:** warehouse portals, smart shelves, conveyor readers, cabinet readers, logistics, high-performance handhelds.

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### 2. Impinj E510 / likely “E518”

I could not confirm a real standard **E518** reader chip, so I am treating this as **E510** unless you confirm otherwise.

The **E510** is the middle-performance E-series chip. It is designed for mobile readers, wearable readers, POS devices, smart appliances, and embedded IoT readers. It gives better performance than E310 but is cheaper/lower power than high-end designs using E710.

Use **E510** when you want:

| Need                   | Good fit? |
|------------------------|-----------|
| Medium read range      | Very good |
| Battery-powered device | Very good |
| POS / mobile reader    | Very good |
| Lower cost than E710   | Good      |

|                     |                                 |
|---------------------|---------------------------------|
| High tag population | Good, but not as strong as E710 |
|---------------------|---------------------------------|

**Best for:** handheld readers, POS readers, wearable RFID devices, compact IoT readers, smart cabinets with medium range.

### 3. Impinj E310

The **E310** is the lower-cost E-series option. It is still a modern chip, but it is designed more for close-range or smaller tag population applications.

Use **E310** when you want:

| Need                    | Good fit? |
|-------------------------|-----------|
| Low cost                | Excellent |
| Short/medium read range | Good      |
| Small group of tags     | Good      |
| High tag population     | Not ideal |
| New design              | Good      |

**Best for:** printers, kiosks, access control, desktop readers, small embedded readers, close-range devices.

### 4. Impinj R2000

The **R2000** was a very popular high-performance older chip. Many UHF RFID modules still use it. It has good sensitivity and high tag read rate, but it consumes more power and uses an older, larger package than the E-series. Impinj's comparison table says **R2000 is not recommended for new designs**.

Use **R2000** when:

| Need                   | Good fit? |
|------------------------|-----------|
| Existing module design | Good      |
| Legacy product support | Good      |

|                                     |                 |
|-------------------------------------|-----------------|
| New product design                  | Not recommended |
| High performance but older platform | Good            |
| Low power                           | Not ideal       |

**Best for:** existing commercial modules, replacement designs, legacy readers, older fixed readers.

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## 5. Impinj R500

The **R500** is the older low-cost Indy-series chip. It is mainly for short-range/proximity RFID readers. It has much lower sensitivity and lower read rate than E710/E510/R2000.

Use **R500** when:

| Need                        | Good fit?       |
|-----------------------------|-----------------|
| Very low-cost legacy design | Good            |
| Short-range reader          | Good            |
| Many tags / long range      | Poor            |
| New product design          | Not recommended |
| Low performance acceptable  | Good            |

**Best for:** older POS readers, proximity readers, smart mirrors, access control, short-range embedded readers.

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## Recommendation

For a **new RFID reader product**, I would choose:

| Application                                         | Recommended chip |
|-----------------------------------------------------|------------------|
| Long-range fixed reader / warehouse gate / conveyor | <b>E710</b>      |
| Handheld reader / battery-powered reader / POS      | <b>E510</b>      |

|                                        |              |
|----------------------------------------|--------------|
| Low-cost close-range embedded reader   | <b>E310</b>  |
| Existing R2000 module replacement only | <b>R2000</b> |
| Existing low-cost legacy reader only   | <b>R500</b>  |

My recommendation: **do not start a new design with R2000 or R500** unless you already have a module, PCB, firmware, or supplier locked to that platform. For new development, use **E710 / E510 / E310**.

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Thought for 4s

Here are the **direct datasheet PDF download links**.

| Chip                       | PDF download link                                                                                                                       |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>E710</b>                | <a href="#">Impinj E910 / E710 / E510 / E310 RAIN RFID Reader Chip Datasheet PDF</a>                                                    |
| <b>E518</b>                | I could not find an official <b>E518</b> chip. It is likely <b>E510</b> . Use this: <a href="#">Impinj E510 RAIN RFID Datasheet PDF</a> |
| <b>E310</b>                | <a href="#">Impinj E910 / E710 / E510 / E310 RAIN RFID Reader Chip Datasheet PDF</a>                                                    |
| <b>R2000</b>               | <a href="#">Impinj Indy R2000 Product Brief PDF</a>                                                                                     |
| <b>R500</b>                | <a href="#">Impinj Indy R500 Product Brief PDF</a>                                                                                      |
| <b>All-chip comparison</b> | <a href="#">Impinj Reader Chip Portfolio Comparison PDF</a>                                                                             |

Also, Impinj's support page confirms the latest public datasheet is for **E910, E710, E510, and E310** reader chips, not E518.